

Lesson 7: THE PLATYHELMINTHES

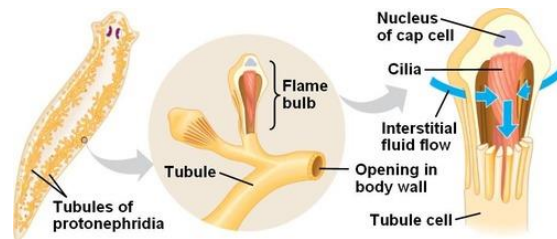
Introduction

- The phylum Platyhelminthes contains triploblastic acoelomate worms with flattened bodies (platys – flat; helmis = worm).
- They are considered the most primitive bilaterally symmetrical animals (bilateria) and the first group to have evolved a true mesoderm and a primitive head.
- All coelomate animals are thought to have evolved from flatworm-like ancestors.

General Characteristics & Adaptive Features of Platyhelminthes

- The phylum Platyhelminthes contains triploblastic acoelomate worms with flattened bodies (platys – flat; helmis = worm).
- The Phylum has about 25000 species.
- Platyhelminths can either be parasitic or free living. They are thought to have evolved from cnidarian ancestors in which the ciliated larva developed into a new adult form with bilateral symmetry.
- They inhabit both aquatic and terrestrial environments.
- This was an adaptation to a more active way of life, with the beginning of cephalization (head formation), with eye-spots, chemical/ touch/pressure sensing cells, and a “brain” (a cluster of nerve cells), at the anterior end.
- Flatworms also have true organs, including layers of muscles that are part of a well developed mesoderm. This layer also contains an excretory system of flame cells and reproductive structures.
- Flatworms retain some ancestral characters; the mouth like in Cnidarians is the digestive tract’s only opening (gut is a blind sac).
- They possess an incomplete digestive tract: a mouth is present, but there is no anus. Digestion is both extracellular and intracellular, and undigested food is expelled through the mouth.
- Some parasitic forms (e.g., tapeworms) lack a digestive system entirely and absorb nutrients directly from the host
- Gas exchange occurs through the general body surface (diffusion across the thin outer body cover); the flattened body shape means that no cell is far from the external source of oxygen.

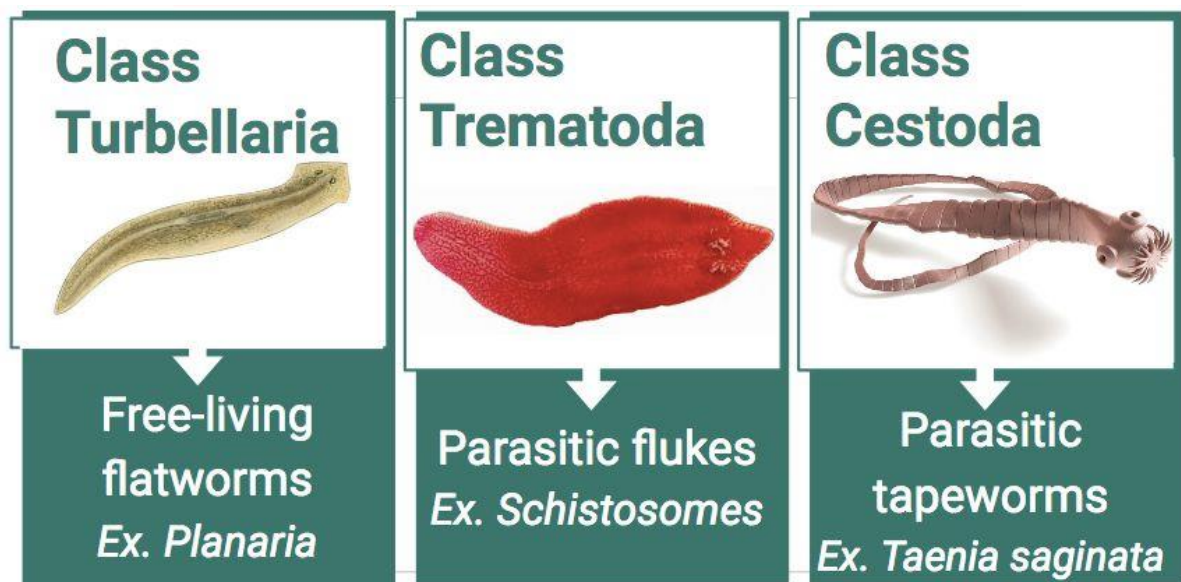
- Excretion occurs through **protonephridia** with flame cells, which help in osmoregulation and removal of waste. This is considered a primitive excretory mechanism compared to nephridia in higher invertebrates.



- Most are **hermaphroditic**, possessing both male and female reproductive organs.
- Fertilization is usually internal, and many exhibit complex life cycles, especially parasitic forms (flukes, tapeworms).

CLASSIFICATION

- The phylum contains three classes of worms.
- ◆ Two of these, the **Trematoda** (flukes) and **Cestoda** (tapeworms) contain only parasites.
- ◆ Class **Turbellaria** (planarians) which contains free living forms.



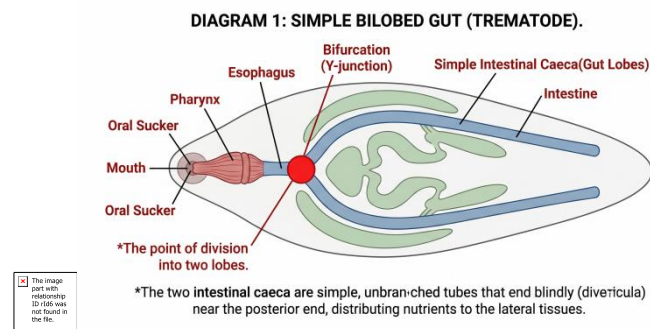
1. Class: Turbellaria

- ✓ Turbellarians are commonly planaria worms or planarians.
- ✓ They are mostly free-living predators and scavengers.
- ✓ They range in size from microscopic size to 60 cm long, although most are less than 1 cm.
- ✓ Cilia cover the entire body in small forms but only on the ventral surface of larger ones.
- ✓ Turbellarians are often recognizable by [light sensing eyespots on the top of the head](#).
- ✓ Most are marine but a few, like *Dugesia* (common planaria worm), live in fresh water.
- ✓ The few terrestrial species spend most of their time in moist areas, under logs and leaf litter.
- ✓ Small-forms that are aquatic have cilia all over their body and use it to swim.
- ✓ Small terrestrial forms move by secreting slimy mucous and gliding along it using ciliary action.
- ✓ The muscles are used for making turns or in curling up when disturbed.
- ✓ Tubellarians feed on smaller organisms or on dead organic matter. Small bits of meat are used as bait to attract them for collection.
- ✓ The first part of the gut (fore gut) is a muscular bulb or tube, which protrudes through the mouth during feeding in order to suck food.
- ✓ Nearly all turbellarians are hermaphroditic.
- ✓ Sexual reproduction occurs by mutual copulation between two individuals, with internal fertilization.
- ✓ After embryonic development, the eggs of some species hatch as miniature adults (direct development), whereas others produce ciliated larvae commonly called Muller's larvae (indirect development).
- ✓ Some turbellarians are [parthenogenetic](#), producing new offsprings from unfertilized eggs, and in some species males are unknown.
- ✓ Many also reproduce asexually by budding.

2. Class – Trematoda

- ✓ They are commonly called flukes .

- ✓ Their **diagnostic feature** is possession of **two adhesive organs or suckers** near the mouth (oral sucker) and a second sucker on the ventral surface (ventral sucker).
- ✓ Their suctorial pharynx sucks host tissues or body fluids in through the mouth and passes them into the simple gut (short intestine).
- ✓ In addition, flukes have a **bilobed gut**.
- ✓ Flukes produce huge numbers of eggs/offsprings, an adaptation to their parasitic way of life.
- ✓ Most flukes are hermaphroditic except (*Schistosoma*-blood flukes) and reproductive organs occupy most of the body.
- ✓ Flukes have complex life cycles involving several different hosts. The host of the adult is the primary/definitive host; hosts of larval stages are intermediate/secondary hosts or vectors.
- ✓ Fluke larvae multiply asexually, and so a single fluke egg can generate many adults. Examples of genera include *Fasciola* (liver flukes), *Paragonimus*.



3. **Class – Cestoda**

- ✓ They are commonly called tapeworms, a group of very specialized parasites that live in the intestines of probably every species of vertebrate.
- ✓ Their adaptation to this way of life involved loss of the head, mouth and digestive system.
- ✓ Always surrounded by digested food, the worm simply absorbs it through the body surface (diffusion).
- ✓ These worms are facultative anaerobes, adapted to their low oxygen habitat.
- ✓ The anterior of the body usually has a scolex ('head') bearing a circle of hooks, suckers or both, which attach them to the lining of the hosts intestine. Behind the

scolex is a growing region (**budding region**) that constantly produces new body sections, the **proglottides**.

- ✓ Although the nervous, muscular and excretory systems, run through-out the chain, a mature proglottid's main features are its reproductive organs (has both ovaries and testes-hermaphroditic).
- ✓ Then full of fertilized eggs, a proglottid breaks away from the rest of the worm (autotomy) and is passed out of the host with the faeces.
- ✓ Many cestode eggs or hatched larva must be eaten by several different host species in order to complete their development.
- ✓ In case they are not eaten, they form dormant cysts in order to withstand unfavourable conditions. Examples of genera include *Taenia*, *Dipyllobothrium*, *Hymenolepis*, *Dipylidium*, *Echinococcus*.

Feature	Turbellaria (Free-living flatworms)	Trematoda (Flukes)	Cestoda (Tapeworms)
Habitat	Free-living in aquatic/terrestrial habitats	Exclusively parasitic (endoparasites in vertebrates)	Endoparasitic in vertebrate intestines
Body covering	Ciliated epidermis for locomotion	Non-ciliated tegument (protective cuticle)	Non-ciliated tegument specialized for absorption
Digestive system	Mouth + pharynx → branched intestine; no anus	Incomplete: mouth → pharynx → bifurcated intestine; no anus	Absent; nutrients absorbed through tegument
Attachment organs	None (free-living)	Suckers and hooks for adhesion	Scolex with suckers/hooks; proglottids for reproduction
Body structure	Soft, unsegmented body	Leaf-like, unsegmented body	Segmented: scolex, neck, strobila (chain of proglottids)
Nervous system	Ladder-type with cerebral ganglia	Reduced, adapted to parasitism	Highly reduced, adapted to parasitism
Reproduction	Hermaphroditic; asexual fission common	Hermaphroditic (except schistosomes); complex life cycles with multiple hosts	Hermaphroditic; each proglottid contains complete reproductive organs
Key distinction	Free-living with cilia	Parasitic with suckers and complex life cycle	Segmented body, no digestive system
Examples	<i>Planaria</i> , <i>Polycladida</i>	<i>Fasciola hepatica</i> , <i>Schistosoma</i>	<i>Taenia solium</i> , <i>Echinococcus</i>